SAFE MIGRATION OF LARGE-SCALE IP/MPLS NETWORKS: A CASE STUDY AND LESSONS LEARNED FROM THE AIRTEL–TIGO GHANA NETWORK CONSOLIDATION

Kwabena Akomea Agyin

IEEE senior member, network engineer, utah, USA

ABSTRACT

Large-scale network modernization can increase capacity, resilience, and readiness for future services, but migrating live services introduces significant operational risk. This study examines the consolidation of two independent IP/MPLS production networks in the Airtel–Tigo Ghana national network, which served approximately 10 million subscribers. The modernization increased key backbone capacity from approximately 30 Gb/s to 100 Gb/s while live services were progressively transferred to the consolidated architecture. The study formalizes a Safe Migration Design Framework based on four operational principles: a migration should be easy to execute, easy to troubleshoot, easy to roll back, and easy to complete. The migration achieved minimal service risk and near-zero disruption while maintaining critical services. The lessons provide a practical framework that network engineers can apply to similar large-scale brownfield network migrations.

KEYWORDS

IP/MPLS, brownfield network migration, network consolidation, safe migration, service continuity, rollback, migration validation, Inter-AS Option A/B/C.

1. INTRODUCTION

Modernizing a production IP/MPLS network introduces a distinct engineering challenge because the infrastructure being upgraded is also carrying live traffic. Changes to routing policies, IP addressing, IGP relationships, MPLS forwarding, VPN services, interface capacity, or redundancy paths can affect service reachability and traffic behaviour. Major service disruptions have occurred when network changes, routing behaviour, implementation scope, testing, or recovery conditions were insufficiently controlled [5], [6]. Migration design must therefore address both the final target architecture and the protection of live services while the network transitions to that state.

Large-scale network consolidation increases this complexity because two production environments may need to coexist during migration. Each network may have its own BGP Autonomous System, routing policies, addressing, MPLS design, data centres, vendor platforms, software versions, and operating procedures. These differences cannot normally be removed through a single cutover. The migration must therefore define both the target architecture and a controlled transition path for moving services from the existing environments into the consolidated network.

Existing standards define MPLS forwarding, MP-BGP extensions, and VPN service architectures [2]–[4], while vendor documentation provides implementation guidance for individual technologies and platforms. Large brownfield migrations, however, require these mechanisms to operate as part of one coordinated engineering process. Network assessment, target design, capacity planning, infrastructure readiness, service dependency analysis, pre-deployment testing, coexistence, migration sequencing, rollback, and validation must support the same migration objective. Practical case-based guidance showing how these activities can be coordinated across a large production IP/MPLS consolidation remains limited [8], [9].

The Airtel–Tigo Ghana network consolidation provides a practical case for examining this problem. Following the merger of Airtel Ghana and Tigo Ghana, the combined operator inherited two independent telecommunications environments [7]. The networks had separate IP/MPLS backbones, BGP Autonomous Systems, routing policies, transport infrastructure, data centres, vendor platforms, and operating procedures. Together, they supported approximately 10 million subscribers and carried voice, mobile data, internet, mobile money, roaming, and enterprise services. Important sections of the existing backbone operated with approximately 30 Gb/s of aggregated capacity, while the modernization introduced a 100GE IP core and consolidated the national backbone and three regional data-centre environments.

The project followed an end-to-end engineering process designed to control migration risk throughout the transformation lifecycle. The existing networks were audited to establish routing states, service dependencies, traffic paths, capacity, infrastructure constraints, and known working conditions. These findings guided the target and transition designs, detailed configurations, pre-deployment testing, coexistence strategy, phased service migration, validation, and rollback planning. Each migration stage was controlled and verified before the network progressed to the next state.

This study formalizes these engineering practices as the Safe Migration Design Framework, organized around four operational principles: a migration should be easy to execute, easy to troubleshoot, easy to roll back, and easy to complete. Together, these principles provide a repeatable way to evaluate whether a migration has clear implementation steps, manageable fault domains, a recoverable previous state, and a defined path from temporary coexistence to the final architecture. The Airtel–Tigo case demonstrates how these principles can be applied to a large brownfield IP/MPLS consolidation while reducing service risk and protecting live services.

A. Research Question

What engineering controls are needed to safely migrate a large-scale brownfield IP/MPLS network while protecting live services?

2. RELATED WORKS

2.1. Network Migration and State Transition

Previous studies have examined network migration from several angles. These include planning, state transition, consistent updates, staged technology adoption, and service-provider transformation. Each area contributes to safe migration, but none covers the full problem addressed in this study.

[8] treats migration as a lifecycle that includes planning, design, implementation, monitoring, and rollback. This supports the view that migration is more than a production cutover. Its focus,

however, is mainly practical implementation. It does not develop a case-based framework for protecting live services throughout a large brownfield IP/MPLS migration.

[9] studies migration as movement from an initial state to a final state through intermediate operations. It shows that the order of changes can affect disruption and network performance. This closely aligns with the present study's focus on existing, intermediate, and final network states. Its scope is narrower, with emphasis on link-weight changes and maintenance scheduling.
[16] focuses on the risks created during network transition. It shows that correct starting and final configurations do not guarantee a safe migration. Intermediate configurations can still produce unintended forwarding behaviour. This supports the need to control the intermediate migration state. The study, however, is centred on consistent updates in a software-defined networking environment.

[17] and [18] examine gradual technology migration in service-provider networks. Both recognise that legacy infrastructure may remain active while new technology is introduced. [17] focuses on the order of SDN upgrades, while [18] focuses on the cost of moving towards software-defined IPv6 networks. These studies support staged migration and coexistence. They give less attention to service continuity, rollback, validation, and recovery during live migration.
[19] examines the transformation of a brownfield service-provider network towards an integrated optical-packet architecture. It covers migration planning, High-Level Design, interoperability, and As-Built implementation. This is closely related to the present study because it uses a real production environment. Its main focus is network transformation, not a common set of controls for execution, troubleshooting, rollback, validation, and completion of live IP/MPLS migration.

Together, [8], [9], and [16]–[19] show that large network migrations are staged transitions that require planning and control of intermediate states. They also show the need for coexistence between legacy and target technologies. A remaining gap is how service continuity, recoverability, validation, and final completion should be controlled together across the full lifecycle of a brownfield IP/MPLS migration. This gap leads directly to the problem of maintaining service reachability while the legacy and target routing domains coexist during migration.

2.2. Inter-AS VPN Connectivity Options

Safe migration of large-scale IP/MPLS networks often requires the legacy and target routing domains to coexist while live services are progressively transferred between them. In such brownfield environments, the method used to exchange VPN reachability across Autonomous System boundaries can directly affect migration complexity, fault isolation, rollback capability, and service continuity. RFC 4364 defines mechanisms for extending MPLS Layer 3 VPN services across multiple Autonomous Systems [4], commonly implemented as Inter-AS Options A, B, and C [15]. These options differ in how VPN routes and labels cross the AS boundary, where VPN routing state is maintained, and how much control-plane coordination is required. Understanding these trade-offs is therefore important when selecting an inter-AS mechanism that supports a controlled and recoverable migration strategy.

In Inter-AS Option A, the Autonomous System Boundary Routers (ASBRs) operate as Provider Edge routers at the boundary between the two Autonomous Systems. A separate Virtual Routing and Forwarding (VRF) instance is maintained for each VPN that must cross the boundary, and the neighbouring ASBR is treated as a Customer Edge peer for that VPN. Routes are exchanged between corresponding VRFs using a PE–CE routing protocol, with eBGP commonly used between the ASBRs. MPLS labels do not need to extend across the inter-AS link because traffic

crosses the boundary using normal IP forwarding. Within each Autonomous System, MP-BGP continues to distribute VPN routes between the local PEs and the ASBR.

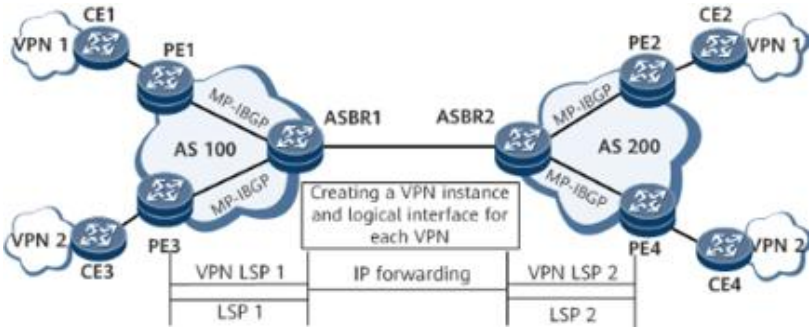


Fig. 1 illustrates the Option A control-plane relationship and VRF-to-VRF service boundary.

The main advantage of Option A is its operational simplicity and strong service separation. Each VPN crossing the boundary can be configured, verified, troubleshot, and removed independently. This also limits the scope of a routing or configuration problem to the affected VRF. Its main limitation is scalability because each additional VPN requires corresponding VRF state and an associated logical or physical interface relationship at the AS boundary. Option A is therefore most suitable where the number of VPNs crossing a specific boundary remains manageable or where operational isolation is more important than large-scale VPN aggregation [15].

In Inter-AS Option B, the ASBRs remove the need for separate VRF-to-VRF connections for every VPN. Instead, the ASBRs establish MP-eBGP sessions and exchange labelled VPN-IPv4 routes directly across the Autonomous System boundary. Each ASBR receives VPN routes from its local network, advertises them to the neighbouring ASBR, updates the BGP next hop as required, and associates the route with the label information needed for forwarding. The inter-AS boundary therefore carries VPN routing and label information directly between the ASBRs.

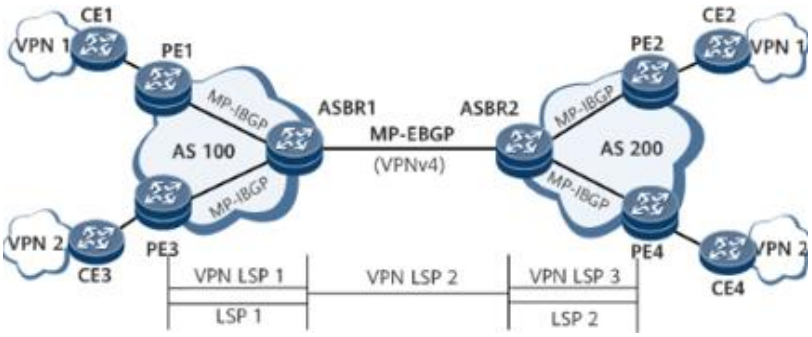


Fig. 2 illustrates the labelled VPN-IPv4 route exchange between the two ASBRs.

Option B provides better VPN scalability than Option A because the boundary is no longer dependent on a separate interface relationship for every VPN. The trade-off is that the ASBRs must maintain VPN routing and label state for the services crossing the boundary. As VPN route volume increases, the control-plane and forwarding-state requirements on the ASBRs also increase. Route-target filtering can reduce unnecessary VPN-route retention, but the ASBRs remain directly involved in VPN route exchange. Option B is therefore appropriate where the scale of VPN connectivity exceeds what is practical with Option A, while maintaining VPN state at the AS boundary remains operationally acceptable [15].

In Inter-AS Option C, VPN-route exchange is separated from the ASBR function. PEs in different Autonomous Systems, or their Route Reflectors, establish multihop MP-eBGP relationships and exchange VPN-IPv4 routes across the participating networks. The ASBRs do not need to maintain the complete VPN-IPv4 routing state. Instead, they exchange labelled IPv4 reachability that supports the transport path required between the ingress and egress PEs.

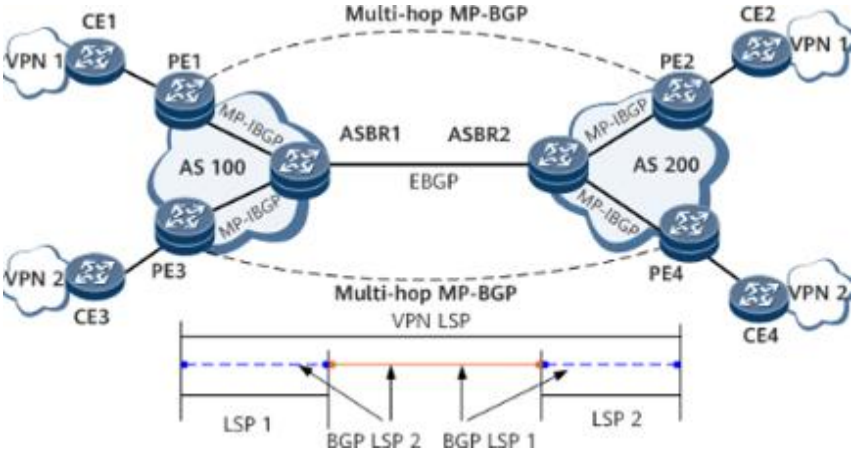


Fig. 3 illustrates the separation between VPN-route exchange and labelled transport reachability in Option C.

This separation provides greater VPN-route scalability because the intermediate ASBRs are not required to maintain every VPN route. VPN routing state remains primarily on the PEs or Route Reflectors, while the ASBRs support labelled transport reachability across the Autonomous System boundary. The trade-off is increased control-plane coordination because end-to-end labelled reachability and multihop VPN-route exchange must be established and maintained across the participating Autonomous Systems [15].

The three methods therefore provide different balances between service isolation, scalability, troubleshooting simplicity, and control-plane complexity. Option A provides a clear per-VRF boundary and simple operational model but becomes less practical as the number of VPNs increases. Option B improves scale by exchanging labelled VPN routes directly between ASBRs, but places VPN routing state on those boundary routers. Option C further separates VPN-route exchange from the ASBR function and provides greater VPN-route scalability, but requires more extensive coordination of labelled reachability across the participating Autonomous Systems.

These differences are particularly relevant to safe brownfield migration because the most scalable inter-AS method is not necessarily the most appropriate temporary migration mechanism. A transition design may favour a simpler and more isolated boundary if its purpose is to support controlled coexistence, limited service movement, rapid troubleshooting, and straightforward removal after migration.

This consideration was central to the Airtel–Tigo consolidation. Inter-AS Option A was used as a temporary site-level coexistence mechanism between the legacy and target routing domains, rather than as the permanent nationwide inter-AS architecture. Only the VRFs required for the services being migrated through a particular local boundary needed to participate at that stage. Its VRF-to-VRF separation therefore allowed each temporary service relationship to be configured, validated, troubleshot, rolled back, and removed independently. In this context, the operational simplicity of Option A aligned directly with the Safe Migration Design Framework and supported a controlled transition between the two production IP/MPLS environments.

2.3. Network Change and Recovery Risk

Previous production-network failures show that changes to backbone routing and configuration can propagate beyond the device being modified and affect multiple dependent services. During the October 2021 Meta outage, backbone configuration changes disrupted communication between data centres and subsequently affected DNS reachability, customer-facing services, and internal operational tools [5]. Similarly, the February 2024 AT&T Mobility outage began during a network change in which an equipment configuration error was introduced; the FCC reported that the resulting disruption lasted at least 12 hours and affected voice, data, and emergency-service connectivity [6]. These incidents demonstrate that migration risk is influenced by service dependencies, implementation scope, and the behaviour of the network while moving between operational states.

The July 2022 Rogers outage further illustrates the relationship between routing changes, control-plane capacity, and recovery capability. An independent assessment published by the Canadian Radio-television and Telecommunications Commission found that removal of an Access Control List policy filter during an IP network update allowed excessive routing information to reach the core routers and exceed their processing capacity [12]. Management connectivity also depended on the affected IP core, which reduced the operator's ability to troubleshoot and restore the network. This case shows that change impact can extend across routing, forwarding, and management functions when these dependencies are not sufficiently isolated.

Prior recovery work also emphasizes the importance of maintaining a known working path and a defined restoration state. The MPLS recovery framework in RFC 3469 distinguishes between the normal working path and the protection or restoration path used when failure occurs [13]. This principle is directly relevant to network migration because a production change temporarily places the network between an existing working state and a target state. Recovery therefore depends on understanding both states and preserving a viable path back to the previous service condition when the transition fails.

Collectively, the reviewed work shows that safe network migration requires staged transition, control of intermediate states, service reachability during coexistence, and recovery planning [4], [5], [6], [8]–[19]. However, these areas are mostly addressed separately. Limited case-based research explains how the existing production state, intermediate migration state, and final target state should be controlled together while live services remain operational. This gap motivates the present study, which examines the engineering controls needed across the full lifecycle of a large brownfield IP/MPLS migration and uses the findings to develop the Safe Migration Design Framework.

3. METHODOLOGY

This study uses the Airtel–Tigo Ghana network consolidation as a production case study to examine how a large-scale brownfield IP/MPLS network can be migrated while protecting live services. The migration lifecycle is analysed against four operational principles: the migration should be easy to execute, easy to troubleshoot, easy to roll back, and easy to complete.

The analysis follows the engineering process used during the project, from customer requirements and network assessment through solution design, controlled coexistence, service migration, validation, recovery, and final network completion. At each stage, the study examines the technical decisions and operational controls used to reduce implementation uncertainty,

contain fault impact, preserve a recoverable working state, and provide a clear path from the existing network to the approved target architecture.

3.1. Customer Requirements for Network Consolidation

Following the Airtel–Tigo merger, the customer issued a Request for Information (RFI) to telecommunications vendors in Ghana to evaluate approaches for consolidating and modernising the two production networks. The RFI defined the business, technical, and service-continuity outcomes that the proposed solution was expected to achieve.

- *Network consolidation:* The proposed solution was required to integrate the existing Airtel and Tigo IP/MPLS, access, data-centre, and supporting network environments into a common national network while maintaining the services already carried by both networks.
- *Infrastructure optimisation:* The customer required opportunities to be identified for consolidating duplicated network functions, equipment, routing roles, licences, site resources, and operational support requirements while preserving the resilience and service capabilities needed by the business.
- *Capacity and technology modernisation:* The consolidated network was expected to support the combined traffic load and projected growth over the planning period. The solution therefore had to provide sufficient backbone capacity, suitable platforms, required software and licences, redundancy, and support for future digital-service requirements.
- *Migration and service continuity:* The customer required the existing services to be transferred into the consolidated network with controlled service impact and continuity maintained throughout the transformation.

3.2. Feasibility Study: Network Audit and Technical Site Survey Framework

A feasibility study was carried out to establish a complete and verified picture of the existing Airtel and Tigo production networks before the target architecture and migration strategy were finalised. The assessment focused on collecting the technical, service, performance, infrastructure, and operational information required to understand how the networks operated, where their limitations existed, and what known working state had to be preserved for validation, troubleshooting, and rollback.

- *Define the audit scope and physical network inventory:* The project engineering team identified the network domains, POPs, data centres, routers, gateways, backbone links, access infrastructure, services, and technical teams within the migration scope. Router roles, interface types, link speeds, available ports, equipment locations, and physical connectivity between sites were then documented. This established what infrastructure existed, where it was located, and which parts of the production environment would be affected by the migration.
- *Document the logical and control-plane environment:* The audit captured the existing BGP Autonomous Systems, OSPF and IS-IS routing domains, MPLS services, VRFs, routing policies, route advertisements, received routes, and route propagation. This established how reachability was maintained across the two production environments and provided the control-plane reference required before any new routing relationship could be introduced.
- *Map every service and traffic flow end to end:* Each service was traced from source to destination. The audit identified the access connection, PE router, VRF, backbone path, gateway, data-centre environment, and other network elements on which the service depended. This established the complete dependency

chain for each service. A single router or backbone link could support several services with different destinations and dependencies, so migration planning had to follow the complete service flow rather than the device alone. This service map later became a key reference for migration sequencing, fault isolation, validation, and rollback.

- *Establish the capacity, performance, and assurance baseline:* Interface capacity, bandwidth utilisation, traffic levels, available headroom, congestion points, latency, packet loss, alarms, routing states, reachability, and other available KPIs were recorded across the network. Important sections of the existing backbone were operating with approximately 30GE maximum capacity. These measurements identified existing capacity and performance limitations and created the known working baseline used later for pre-change, post-change, and rollback validation.
- *Assess platforms, software, licences, lifecycle, and support readiness:* Device vendors, hardware platforms, software releases, installed licences, supported features, lifecycle status, and vendor roadmaps were documented. Existing support contracts, escalation matrices, technical contacts, expected response times, Huawei TAC, R&D, customer specialists, and other support resources were also reviewed. This established what technology could remain in service, what limitations had to be addressed, and what specialist support would be available if issues arose during migration.
- *Conduct Technical Site Surveys and determine site readiness:* POPs, data centres, and colocation facilities were physically inspected for rack space, power availability, cooling, grounding, monitoring systems, fibre routes, cable lengths, ducts, connectivity points, equipment locations, and available router interfaces. Required rack, power, fibre, and cabling preparation was documented before installation. The surveys also identified the cross-functional teams responsible for site access, fibre work, equipment installation, testing, handoffs, and operational support.
- *Review and reconcile the audit with the live network:* The collected findings were reviewed with Airtel and Tigo network engineers, Huawei TAC, R&D, and other technical specialists involved in the migration. The topology, routing state, service flows, capacity, platform information, and site conditions were checked against the actual production environment. Where existing documentation differed from the verified live state, the live network was used as the reference and the audit was corrected.
- *Approve and maintain the production baseline:* After technical validation and sign-off, the audit became the agreed production-network snapshot. Any subsequent change involving services, routing, configurations, capacity, interfaces, or infrastructure was incorporated before the affected migration window. This ensured that design, troubleshooting, validation, and rollback decisions continued to use the actual network state at the time of migration.

a. Key Findings from the Feasibility Study

The feasibility study identified several constraints that directly shaped the migration strategy:

- *Limited spare backbone fibre:* One of the most significant physical findings was the limited or unavailable spare duct fibre cores between some POPs and data centres. Existing fibre cores were already carrying production backbone traffic, which meant that a completely independent parallel backbone could not be built at all locations before migration.
- *Limited spare interfaces on legacy routers:* Some existing routers had limited free ports available for temporary interconnection with the new equipment. This restricted the amount of old-to-new parallel connectivity that could

be established during the coexistence period and required the migration design to make efficient use of the available interfaces.

- *Performance and capacity limitations:* The audit identified performance limitations from the collected KPI baseline, including latency, packet-loss, throughput, and capacity constraints on parts of the existing network as traffic demand continued to grow. Some backbone sections were operating with limited bandwidth headroom, which increased the importance of carefully dimensioning both the target network and any temporary migration paths used during coexistence.
- *Redundancy mechanisms approaching scalability limits:* The assessment also identified areas where the existing redundancy and failover mechanisms were becoming less suitable as network scale and traffic capacity increased. The target design therefore had to provide stronger path diversity, faster failure detection, and faster traffic recovery so that resilience could scale with the modernised backbone.

3.3. Solution Design Phase: HLD, Topology, LLD, Configuration Development, and MOP Framework

The solution-design phase converted the verified production baseline into an architecture and migration process that could be implemented on the live network. The work progressed from the High-Level Design (HLD) through detailed topology and Low-Level Design (LLD) to configurations, testing, and the final Method of Procedure (MOP). Each stage progressively reduced uncertainty before production traffic was exposed to change.

- *Develop the High-Level Design:* The HLD was developed from the customer requirements, audit findings, Technical Site Survey, identified network limitations, budget, project delivery timeframe, and future traffic forecast. It defined the target backbone architecture, capacity, routing approach, redundancy, service requirements, technology roadmap, and overall transition from the Airtel and Tigo environments to the consolidated network.
- *Plan for future service demand:* Platform sizing considered a minimum five-year traffic-growth horizon and the future technology roadmap. The target design increased key backbone capacity from approximately **30GE to 100GE** and incorporated capabilities required for IPv6 and future 5G services.
- *Select hardware, software, licences, and support:* Hardware and software were assessed against forwarding capacity, interface density, redundancy, optical compatibility, MPLS capability, IPv6 support, routing scale, lifecycle, and licence requirements. TAC and R&D availability, support response times, software maintenance, and spare availability were also considered so that the selected platform remained supportable during deployment and operation.
- *Develop the physical and logical topology:* The HLD was translated into the target arrangement of P routers, PE routers, Route Reflectors, gateways, access functions, data-centre environments, backbone links, and redundancy relationships. This defined the intended final forwarding structure before implementation details were developed.
- *Design the backbone routing architecture:* IS-IS was introduced as the backbone Interior Gateway Protocol and engineered to support fast convergence. Adjacencies, metrics, path diversity, and redundancy were designed to allow traffic to recover rapidly when a link or network node became unavailable.
- *Improve failure detection and traffic recovery:* Bidirectional Forwarding Detection (BFD) was introduced to shorten failure-detection time.

Fast-reroute capabilities were also incorporated with the service-routing design so that alternate paths could be available when a preferred path failed. These mechanisms reduced the period during which a transport failure could affect service forwarding.

- *Introduce IPv6 capability:* IPv6 addressing, routing support, platform capability, and coexistence with IPv4 were incorporated into the new architecture. Including IPv6 during the modernization avoided another major redesign when future services required IPv6 transport.
- *Design the transition architecture:* The engineering team defined how the network would move from the existing state through temporary coexistence and into the target architecture. Temporary connectivity, legacy-to-new routing relationships, intermediate traffic paths, and removal criteria were therefore designed before migration began.
- *Develop the Low-Level Design:* The approved HLD was expanded into detailed fibre connectivity, port maps, interface rates, addressing, IS-IS adjacencies, BGP and MP-BGP relationships, MPLS services, VRFs, redundancy, service interfaces, and migration-specific connections.
- *Develop the service migration design:* The service-flow information collected during the audit was used to define the existing path, temporary migration path, target path, and recovery path for each service. This ensured that service movement was designed around end-to-end dependencies instead of individual devices.
- *Convert the LLD into deployment configurations:* The approved design was translated into device configurations, templates, playbooks, and automated scripts where appropriate. CLI or model-driven methods such as NETCONF/YANG could be used depending on platform support.
- *Pre-test the proposed solution:* Configurations and routing behaviour were tested in customer laboratories or vendor test beds using real equipment where available. Testing addressed interfaces, addressing, routing adjacencies, VRFs, route exchange, redundancy, interoperability, and service reachability. Problems identified during testing were corrected before production deployment.
- *Develop the Method of Procedure:* Each migration activity was converted into an operational MOP. The MOP specified the action, owner, execution time, duration, expected result, validation method, troubleshooting allowance, rollback trigger, rollback procedure, recovery owner, escalation path, contingency plan, and decision authority.
- *Define pre-change and post-change checks:* Routing adjacencies, MPLS state, VRF routes, interface status, utilisation, traffic levels, latency, packet loss, alarms, and service reachability were captured before each change. The corresponding post-change checks were defined in advance so that the resulting state could be compared with both the baseline and the intended target state.
- *Apply multi-level technical approval:* The proposed solution was reviewed through customer approval, Huawei TAC and R&D approval, and Project Management approval. The completed technical solution and MOP were then presented to the Change Management Board before the production window.

In summary, the solution-design phase converted the verified baseline into a reviewed, tested, measurable, and recoverable migration solution. Prepared configurations and deterministic MOP steps supported execution, defined expected states supported troubleshooting, predefined recovery procedures supported rollback, and the transition architecture established a path to final completion.

3.4. Site-to-Site Migration Strategy Framework

The migration strategy was developed specifically around the brownfield constraints identified during the feasibility study. Limited spare backbone fibre and limited free interfaces on some legacy routers made a complete parallel build impractical. The project therefore adopted a site-to-site migration strategy in which services were progressively moved onto the new routers while the existing backbone remained available until the replacement path had been prepared and validated.

- *Separate UNI and NNI migration functions:* Each legacy router was logically divided into the User-Network Interface (UNI) and Network-to-Network Interface (NNI) functions. UNI represented the service-facing connectivity, while NNI represented backbone-facing transport. This allowed service migration to be separated from backbone migration.
- *Install and prepare the new router:* The new router was installed beside the existing router and commissioned before live service movement. Required interfaces, VRFs, routing policies, and service configurations were prepared in advance.
- *Establish temporary local connectivity:* A temporary 10GE old-to-new connection was created so that services moved to the new router could continue using the existing backbone while the production backbone fibre remained connected to the legacy equipment.
- *Establish Inter-AS Option A coexistence:* During the transition, the legacy and target networks operated as separate routing domains. Inter-AS Option A was established across the temporary connection using a VRF-to-VRF model. This mechanism was selected because individual VRFs could be extended, tested, troubleshot, and removed independently without extending the complete VPN control plane across the temporary boundary. The simplicity and service isolation of Option A therefore suited the temporary coexistence requirement.
- *Validate the temporary path:* Point-to-point reachability, eBGP adjacency, and expected route exchange within each participating VRF were confirmed before production services were moved.
- *Migrate the UNI services first:* UNI services were progressively transferred to the new router while their NNI traffic returned through the temporary Option A path to the legacy router and then entered the existing backbone. This allowed the service-facing portion of the router to move without simultaneously changing the backbone transport path.
- *Scale the temporary interconnection:* Additional physical interfaces could be added to a LAG as migrated traffic increased. This provided sufficient coexistence capacity and reduced the likelihood that the temporary path itself would become a migration bottleneck.
- *Reduce the legacy router to the NNI function:* Once the UNI services had moved, the legacy router remained primarily because the production backbone fibre still terminated on it. This created a controlled intermediate state before the backbone connection was transferred.
- *Repeat the process at the peer POP:* The same coexistence and UNI migration procedure was completed at the opposite end of the backbone link before the shared production fibre was moved.
- *Apply an NNI readiness gate:* Backbone cutover proceeded only after both new endpoints were commissioned, UNI

services were stable, coexistence paths were operational, capacity was adequate, target 100GE NNI configurations had been approved, and rollback remained available.

- *Move the backbone fibre to the new 100GE NNI:* The existing backbone fibre was then transferred during an approved maintenance window. Before cutover: New Router A → Option A → Legacy Router A → Existing Backbone → Legacy Router B → Option A → New Router B After cutover: New Router A → 100GE Backbone → New Router B
- *Validate the target NNI:* Interface state, IS-IS adjacency, BFD status, expected routes, applicable MPLS forwarding state, traffic levels, alarms, and end-to-end service reachability were checked before acceptance.
- *Preserve rollback until acceptance:* The previous physical and logical state remained available until the new UNI or NNI path passed validation. A failed migration could therefore be reversed directly to the previously verified path.
- *Remove temporary coexistence after success:* Option A relationships, temporary LAGs, and migration-specific connectivity were removed once the new routers had assumed both UNI and NNI functions. This provided a defined end to the temporary migration architecture.

The site-to-site framework implemented the four principles directly. It kept each change small enough to execute and troubleshoot, preserved the previous working path for rollback, and provided a clear progression from temporary coexistence to the final backbone. To support reproducibility, Fig. 4 shows a representative replica of the migration topology. It preserves the key routing relationships, protocol roles, coexistence design, and UNI/NNI migration sequence used in the project. For confidentiality reasons, it does not reproduce the customer's exact production topology, site names, addressing structure, node count, or full interconnection details. However, it retains enough technical detail to support understanding and reproducibility of the migration solution.

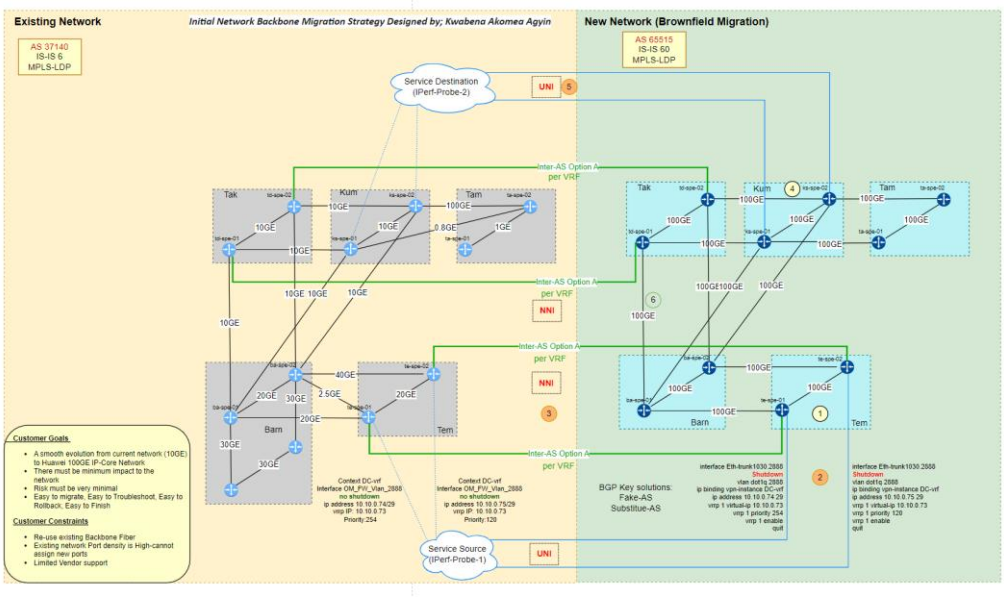


Fig. 4. Representative Airtel–Tigo Brownfield IP/MPLS Backbone Migration and Coexistence Topology

3.5. Service Migration, Validation, and Recovery Decision Framework

The service migration phase determined whether each technical change had actually produced the intended service state. Migration progression was therefore based on measured network behaviour rather than completion of the configuration or physical move alone.

- *Limit each migration scope:* Each window covered a defined set of sites, interfaces, VRFs, links, or services. This restricted the number of variables changed at one time and reduced the fault domain if an unexpected condition occurred.
- *Confirm migration readiness:* The approved MOP, configurations, scripts, support resources, target interfaces, temporary paths, and rollback procedures were checked before execution.
- *Verify the current working state:* The latest routing, forwarding, traffic, alarm, and service indicators were reviewed immediately before the change and compared with the approved baseline.
- *Execute according to the MOP:* Actions were performed in the approved sequence and stopped at defined validation points before the next activity began.
- *Validate the affected service path:* After each move, routing adjacencies, VRF routes, forwarding paths, traffic behaviour, alarms, and service reachability were checked for the specific migration scope.
- *Compare pre-change, expected, and observed states:* The post-change condition was compared with both the original baseline and the target design. Differences outside the approved criteria triggered investigation before progression.
- *Apply the decision gate:* A migration progressed only when the expected result was verified. An uncertain state was held for troubleshooting, while failure against defined acceptance criteria triggered recovery.
- *Use the predefined escalation path:* Issues exceeding the planned troubleshooting scope were escalated through the customer, Huawei TAC, R&D, and other support resources already identified before the change.
- *Restore the known working state when required:* Where rollback was triggered, the previous physical or logical path was restored and checked against the approved baseline before the migration activity was closed.

3.6. Pre- and Post-Migration Transport Performance Validation

To assess the success of the project, key transport performance indicators were compared before and after migration. Measurements were taken across a representative long-distance transport path between the farthest sampled network endpoints. The assessment covered latency, jitter, throughput, and packet loss. Latency was measured using ICMP echorequests, throughput was measured using TCP traffic generated with iPerf, and packet loss was measured using UDP traffic generated with iPerf. Jitter was derived from the variation in repeated latency measurements. Each pre- and post-migration test included 100 measurement samples under comparable test conditions.

a. Latency and Jitter Analysis

Latency represented the round-trip delay across the transport path, while jitter represented variation in packet delay.

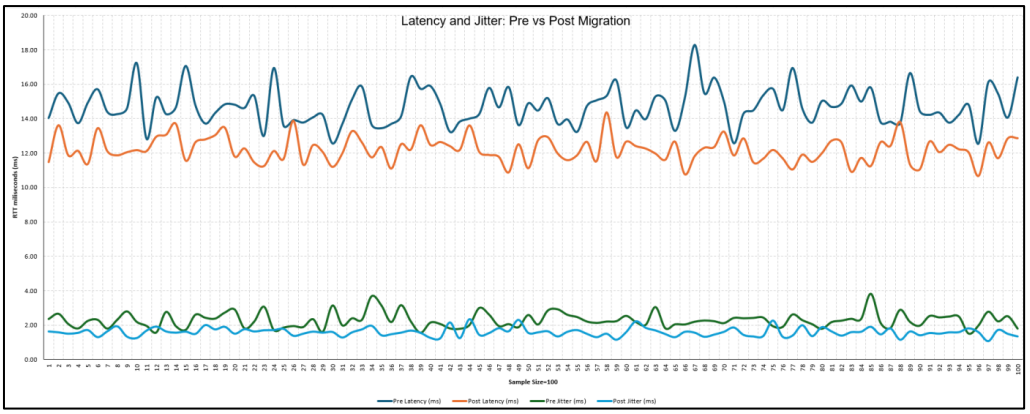


Fig. 5. Latency and Jitter distribution over 100 samples for pre- and post-migration

Before migration, average latency was 14.7 ms and jitter was 2.3 ms. After migration, latency decreased to 12.2 ms, while jitter decreased to 1.6 ms. This represented a 17.0% reduction in latency and a 30.4% reduction in jitter. The results indicated lower delay and more consistent packet delivery after migration.

b. Throughput and Capacity Analysis

Throughput represented the effective data rate successfully carried across the tested transport path. It was measured using sustained TCP traffic generated with iPerf.

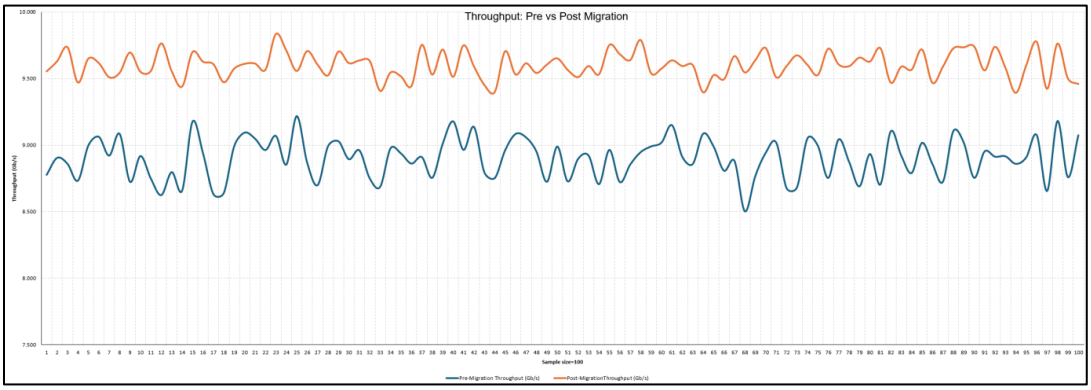


Fig. 6. Throughput distribution over 100 samples pre- and post-migration validation

Measured throughput increased from 8.9 Gb/s before migration to 9.6 Gb/s after migration, representing an increase of approximately 7.9%. The result showed that the migrated path maintained strong data-transfer performance. Backbone capacity increased separately from approximately 30 Gb/s to 100 Gb/s, representing a 233% increase. The target network had been designed to support the combined traffic load and projected growth over the planning period.

c. Packet Loss Analysis

Packet loss represented the percentage of transmitted packets that failed to reach their destination. It was measured using UDP traffic generated with iPerf.

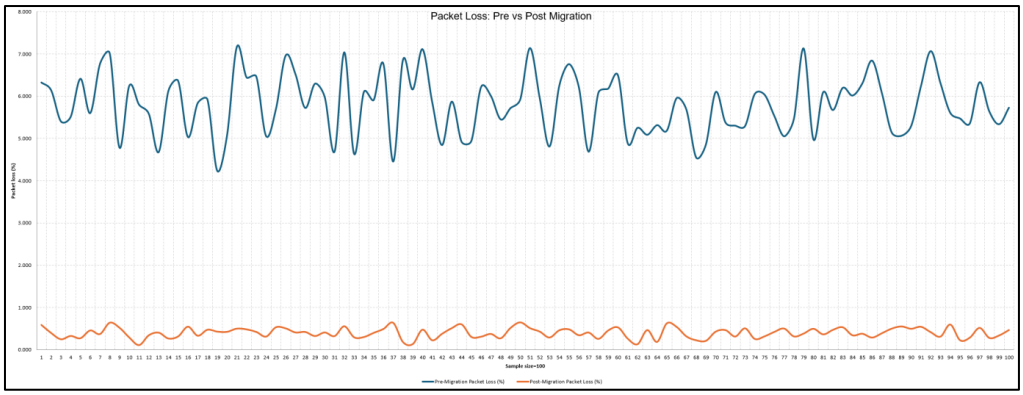


Fig. 7. Packet loss (%) distribution over 100 samples pre- and post-migration validation

Packet loss decreased from approximately 5.8% before migration to 0.4% after migration. This represented a reduction of 5.4 percentage points, or approximately 93.1%.The lower post-migration value indicated improved packet-delivery reliability across the migrated transport path.

3.7. Migration Completion, As-Built Validation, and Handover Framework

The final stage confirmed that successful service movement had resulted in the approved production architecture. Completion therefore required removal of temporary migration mechanisms, verification of the final network state, and transfer of accurate operational information to the teams responsible for the network.

- *Confirm final service placement:* Services were verified to be using the intended target routers, VRFs, backbone links, gateways, and data-centre paths.
- *Remove temporary coexistence mechanisms:* Inter-AS Option A relationships, temporary LAGs, old-to-new connections, and other migration-specific dependencies were removed once their purpose had been completed.
- *Verify the final routing and forwarding state:* Routing adjacencies, BGP relationships, MPLS forwarding, VRFs, interface utilisation, redundancy, alarms, and service reachability were checked after temporary dependencies had been removed.
- *Confirm that legacy dependencies no longer existed:* Service paths were reviewed to ensure that traffic was no longer relying on obsolete routers, temporary VRFs, old links, or migration-specific routing relationships.
- *Decommission legacy equipment after acceptance:* Legacy devices were removed from active service only after their required functions had been transferred and validated on the target infrastructure.
- *Update the As-Built documentation:* The final topology, fibre connectivity, router ports, addressing, routing relationships, VRFs, service paths, and configurations were updated to represent the network as actually implemented.
- *Complete operational handover:* The completed infrastructure, technical documentation, procedures, and relevant knowledge were transferred to the customer operations and support teams.
- *Obtain final acceptance:* Technical and customer acceptance established the point at which migration activity ended and normal production operation of the consolidated network began.

4. DISCUSSION

The Airtel–Tigo case indicates that safe migration of a large production IP/MPLS network is shaped by engineering decisions made throughout the transformation lifecycle. This study therefore examined the research question: What engineering controls are needed to safely migrate a large-scale brownfield IP/MPLS network while protecting live services? The findings showed that these controls must address three main network states: the existing production state, the intermediate migration state, and the final target state.

The existing production state provided the starting point for the migration. Before any major change was made, the engineering team established how the live network was operating. This included the physical topology, routing state, capacity, performance conditions, and service dependencies. The findings were checked against the actual production network. This baseline made it possible to identify what was changing and which services could be affected. It also defined the working state that had to be protected or restored if a migration step failed. Service dependencies were important because a single router could support several VRFs, gateways, access connections, and destination paths. Migration planning therefore had to follow the complete service path, not the device alone.

The intermediate migration state covered the period when the legacy and target networks operated together. Limited spare fibre and router interfaces meant that a fully separate parallel backbone could not be built at every location. The migration therefore used controlled coexistence. UNI services were moved in stages while the existing backbone remained available. Temporary Inter-AS Option A connectivity maintained service reachability between the legacy and target routing domains. The previous working path remained available until the new path had been prepared and validated. Each migration step also had defined checks, rollback triggers, and recovery procedures. Progress continued only after the expected network and service conditions were confirmed. This showed that the intermediate state had to be deliberately designed, limited in scope, and recoverable.

The final target state represented the point at which the migrated network had to be verified against the intended production architecture. Pre- and post-migration measurements were used to assess the migrated transport path. The results showed lower latency, jitter, and packet loss, together with higher throughput after migration. Key backbone capacity also increased from approximately 30 Gb/s to 100 Gb/s. After successful service migration, temporary Inter-AS relationships, migration links, and legacy dependencies were removed. The final routing and forwarding state was verified, As-Built documentation was updated, and the network was handed over for normal operation. Migration was therefore completed only when the temporary transition state had been removed and the target network had become the verified production state.

5. CONCLUSION

This study identified the engineering controls needed to safely migrate a large-scale brownfield IP/MPLS network while protecting live services. The Airtel–Tigo case showed that these controls must span the existing production state, the intermediate migration state, and the final target state. They included a verified production baseline, clear service dependencies, controlled coexistence, preserved rollback capability, predefined validation and recovery conditions, and verification of the final production state.

These lessons were synthesised into the Safe Migration Design Framework, based on four principles: easy to execute, easy to troubleshoot, easy to roll back, and easy to complete.

Together, the principles provide a practical way to assess whether a migration has the controls required to move safely from the existing network, through temporary coexistence, to the final production architecture.

In the Airtel–Tigo case, key backbone capacity increased from approximately 30 Gb/s to 100 Gb/s while critical services were maintained. The migration achieved near-zero service disruption and zero outages outside approved maintenance windows.The main contribution of this study is a practical, case-based framework for designing and evaluating large-scale brownfield IP/MPLS migrations.

6. FUTURE WORKS

Future work should focus on extending and validating the Safe Migration Design Framework in three areas:

- Validation across additional production networks. Apply the framework to other large-scale network migrations involving different vendors, network sizes, service environments, and routing architectures. This will help determine how consistently the four principles apply beyond the Airtel–Tigo case.
- Application to SRv6 migration architectures. Evaluate how the framework performs when service-provider cores evolve from traditional MPLS architectures to Segment Routing over IPv6 (SRv6). Future studies should examine how the four principles support safe coexistence, service migration, rollback, validation, and final transition to an SRv6-based core.
- Automated and AI-assisted migration assurance. Extend the framework with automation and AI techniques that can analyse network baselines, service dependencies, migration states, validation criteria, and recovery plans. This could support migration-readiness checks, detect missing controls, and improve decision-making during large-scale network transformations.

REFERENCES

[1] GSMA, *The Mobile Economy Africa 2026*. London, U.K.: GSM Association, 2026.
[2] E. Rosen, A. Viswanathan, and R. Callon, “Multiprotocol Label Switching architecture,” Internet Engineering Task Force, RFC 3031, Jan. 2001.
[3] T. Bates, R. Chandra, D. Katz, and Y. Rekhter, “Multiprotocol extensions for BGP-4,” Internet Engineering Task Force, RFC 4760, Jan. 2007.
[4] E. Rosen and Y. Rekhter, “BGP/MPLS IP virtual private networks (VPNs),” Internet Engineering Task Force, RFC 4364, Feb. 2006.
[5] Meta Engineering, “Update about the October 4th outage,” Meta, Oct. 2021. [Online]. Accessed: Aug. 8, 2026.
[6] Federal Communications Commission, *February 22, 2024 AT&T Mobility Network Outage: Report and Findings*. Washington, DC, USA: FCC, 2024.
[7] Ghana Chamber of Telecommunications, “Airtel and Tigo merge to form AirtelTigo,” Accra, Ghana, 2017. [Online]. Accessed: Aug. 8, 2026.
[8] G. Gómez Herrero and J. A. Bernal van der Ven, *Network Mergers and Migrations: Junos Design and Implementation*. Chichester, U.K.: Wiley, 2010, doi: 10.1002/9780470660164.
[9] S. Raza, Y. Zhu, and C.-N. Chuah, “Graceful network state migrations,” *IEEE/ACM Trans. Netw.*, vol. 19, no. 4, pp. 1097–1110, Aug. 2011, doi: 10.1109/TNET.2010.2097604.
[10] Cisco Systems, “Configuring MPLS VPN InterAS options,” *Cisco IOS XE MPLS Configuration Guide*. [Online]. Accessed: Aug. 8, 2026.
[11] Juniper Networks, “Interprovider and carrier-of-carriers VPNs,” *Junos OS Documentation*. [Online]. Accessed: Aug. 8, 2026.

[12] Canadian Radio-television and Telecommunications Commission, *Assessment of Rogers Networks for Resiliency and Reliability Following the 8 July 2022 Outage*, Jul. 2024.
[13] V. Sharma and F. Hellstrand, "Framework for MPLS-based recovery," Internet Engineering Task Force, RFC 3469, Feb. 2003.
[14] Huawei Technologies Co., Ltd., "Evolution from traditional VPN to EVPN," *IP Network Series Evolution Guide*, Apr. 2025. [Online].
[15] Huawei Technologies Co., Ltd., "Inter-AS VPN," *Huawei HedEx Product Documentation*, Doc. EDOC1100331384. [Online]. Available: https://support.huawei.com/hedex/hdx.do?docid=EDOC1100331384&id=EN-US_CONCEPT_0000001090123592. Accessed: Aug. 9, 2026.
[16] M. Reitblatt, N. Foster, J. Rexford, C. Schlesinger, and D. Walker, "Abstractions for network update," in *Proc. ACM SIGCOMM 2012*, Helsinki, Finland, 2012, pp. 323–334, doi: 10.1145/2342356.2342427.
[17] K. Poularakis, G. Iosifidis, G. Smaragdakis, and L. Tassiulas, "One step at a time: Optimizing SDN upgrades in ISP networks," in *Proc. IEEE INFOCOM 2017*, Atlanta, GA, USA, 2017, pp. 1–9, doi: 10.1109/INFOCOM.2017.8057136.
[18] B. R. Dawadi, D. B. Rawat, S. R. Joshi, P. Manzoni, and M. M. Keitsch, "Migration cost optimization for service provider legacy network migration to software-defined IPv6 network," *Int. J. Netw. Manage.*, vol. 31, no. 4, Art. no. e2145, Jul. 2021, doi: 10.1002/nem.2145.
[19] N. Haddaji, K.-K. Nguyen, and M. Cheriet, "Towards end-to-end integrated optical packet network: Empirical analysis," *Opt. Switching Netw.*, vol. 27, pp. 18–39, Jan. 2018, doi: 10.1016/j.osn.2017.06.003.

AUTHOR



Kwabena Akomea Agyin is a network engineer and researcher with extensive experience in large-scale telecommunications modernization and safe network migration. He has supported major transformation programs for MTN Ghana, Vodafone Ghana, AirtelTigo Ghana, MTN Liberia, AFCOM Sierra Leone, Ghana's National Interconnect Clearinghouse, and Rural Telephony Project. His work spans IP/MPLS backbone consolidation, IP-Core and IP-RAN modernization, service migration, live-network cutovers, capacity expansion, and SRv6 deployment. His research focuses on safe migration strategies that protect services while enabling future-ready network evolution.